

Intelligent Optimal Framework for the Industrial Mining Plant-Wide Prediction Control

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Abstract—In the underground mining industry, the deep cone system (DCS) is the core element for the whole production and paste-filling process. Due to the coupled variables, large time delay, and the harsh environment, it is critical to developing a framework for the key-quality parameters control in the cone thickener system (CTS). However, the current method cannot finish the high-quality criteria for the intelligent mining paste. Furthermore, the prediction and control are different to fully considered in a unified framework. To address these issues, this article proposes a new unified optimal prediction control framework (ConFrame) for the industrial mining system. The main implementation is divided into two stages, 1) the prior prediction model is built for the key-quality forecasting in the time series and 2) a complete prediction model is then used as the core objective function for accomplishing the optimal ConFrame and parameter solution. The proposed framework accomplishes the integration of optimal and nonlinear control for the industrial paste-filling system. In addition, the presented framework narrows the gap between the mining concentration prediction and optimal control and provides a unified consideration for the full-scale pasting-filling process. Furthermore, the approximation ability and convergence of the proposed algorithm are theoretically analyzed. Finally, the comparative performance is carried out on the numerical Rastrigin benchmark case to highlight the efficacy of the presented framework over the state-of-the-art framework. The presented framework is further tested in the practical cone thickener process which presents the prediction and control performance with high accuracy and robustness.

Index Terms—Bidirectional long-short time memory (BiLSTM), deep cone thickener, improved particle swarm optimization (PSO), optimal prediction control, process modeling.

NOMENCLATURE

Number Sets

- $\mathbb{C}$ Complex numbers.
- $\mathbb{F}$ Mud flow height.
- $\mathbb{H}$ Mud height.
- $\mathbb{L}$ Number of layers.
- $\mathbb{R}$ Real numbers.
- $\mathbb{W}$ Weights of Matrix.

Other Symbols

- Δt Time delay.
- $\hat{y}_i$ Prediction of underflow concentration.
- y_i Actual value of underflow concentration.
- RNN Recurrent neural network.
- GRU Gated recurrent unit.
- ρ Friction index.
- DCS Deep cone system.
- DCT Deep cone thickener.
- f Nonlinear prediction mapping function.
- LSTM Long short-time memory.
- V Constant volume.

Physics Constants

- C_1 Feed concentration.
- C_2 Underflow concentration.
- k Flexible spring coefficient.
- P Mud pressure.
- Q_1 Feed flow amount.
- Q_2 Underflow amount.
- s Efficiency coefficient.

I. INTRODUCTION

THE underground mining industry is the pillar of the national and civilization economy, which supports the development of science and technology [1], [2], [3]. The pasting filling is the main process for the exploitation and utilization of underground resources. Nevertheless, the traditional mining industry faces the difficulty of low-carbon, clean production requirements, which means the revolution of smart mining is still underway. The goal of mining is to realize autonomous high-efficient mining without manual interrupt and operation. However, the traditional mining industry faces challenges with high manual cost, low-efficient operation, and high safety hazards. In addition, the conventional pasting filling process is controlled by the closed-circle method, which means the device and manual cost is so high, further, the manual operation in the deep mining bath is dangerous for the mining workers. The harsh environment in the underground makes it needs plant-wide control for intelligent mining industry employment. With the development of deep learning and smart manufacturing, the pasting-filling process of the mine systems is facing a revolution with advanced control [4].

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Nowadays, the mining industry-based frameworks can be classified into two categories. The first category is model-based, which needs an accurate mathematical model of the reaction process [5]. On the other side, some data-driven model-free methods have been widely applied to the mining pasting process. For example, Jiao et al. [1] developed the solid concentration distribution in the deep cone thickener. They contribute to the geometry of the DCS based on a pilot scale test. Chen et al. [2] then proposed pore connectivity and dewatering tailings and running mechanisms of the raking distortion in the DCS system. As for the underflow concentration prediction methods, Yuan et al. [3] presented a dual-attention recurrent neural network method for deep cone thickener application. The performance of their network-based method has been verified in the real DCS system. The prediction accuracy is high and the model is efficient, however, the final objective of smart mining is to realize unified prediction and control. Their methods ignore the control of the underflow concentration stability and control of the DCS. The bidirectional method motivates us to build a higher-performance of prediction model with lower MSE and higher robustness and stability. Ruan et al. [4] investigated some numerical aspects of the flocculation and settling behavior of the whole-tailings particles in the deep-cone thickener. However, the variant of the DCS is various and the generation of the numerical simulation cannot be directly applied to the industrial case.

Some researchers also considered the control of the paste-filling process, as shown in [6], they used a particle swarm optimization (PSO) algorithm with a single feedforward neural network to forecast the strength of paste-filling material. Xu et al. [8] studied the technology of partial filling to control coal mining subsidence. Wang et al. [11] presented an industrial paste-filling weighting control system and gave us a basic optimization with an artificial network. Wu et al. [12] simulated the long-distance pipeline transportation properties of the whole-tailings paste with a high sliming approach [12]. Their limitation is the lack of an integrated framework that realizes the autonomous intelligent mining framework.

In the last decades, there are also many frameworks for the industrial process. However, there are some apparent shortcomings existed in them. The first factor is that the system is single and cannot run on the plant-wide level. The system achieves high efficiency in the basic unit while decreasing a lot when run in the large-scale process. The second drawback is the complex computation cost of the deployment of these distributed central systems. At last, the multiple tasks cooperate are hard to realize and the generation and robustness are hard to guarantee, which means it is not available to transfer the model to other application processes. Therefore, the intelligence of the mining system and the paste-filling process still needs to be improved.

To sum up, the existing platform is just some part of prediction or the control system in the mining paste-filling industry, there is no unified intelligent system and platform which implements the multiple parameter prediction task and the coordinate control, which motivates the purpose of the novel framework for promoting the development of mining industry and farsightedness.

Based on the above-mentioned analysis, we therein present a control framework (ConFrame) for the industrial mining process. The historical information is considered in the proposed method, an improved PSO method is also given in a detailed way. To sum up, the main contributions of this article are as follows.

- 1) To tackle the hardship of intelligent mining, we present a full-scale framework or system platform for the underground mining industry. The presented framework considers the prediction and optimization problem in the pasting fill process. Distinct from the traditional LSTM [27], [28], [29], [37], the proposed prediction method developed dual feedforward and inverse bidirectional LSTM to learn complex sequence features and improve prediction accuracy.
- 2) Compared with the traditional framework [30], [32], [34], [36], the presented ConFrame addresses the prediction and optimization issues in the underground mining system, and a novel uninformed consensus is given. The specific advanced underflow concentration algorithm is employed in the proposed platform, and the further optimization method is then inherited.
- 3) The advanced optimization for the key-quality control parameter is given for the deep cone thicker system, which differs from the traditional PSO [16], [28], [29], [31], an improved PSO algorithm is designed for solving the industrial mining optimization control. The global intelligent platform makes it more convenient for the target of “one-key” pasting for the mining industry.
- 4) Unlike the previous mainstream frameworks (Table I), the advised ConFrame implemented the unified prediction and control tasks based on the data-centric model, the real-time prediction and control are given, so the operation cost for the mining pasting process is greatly reduced. On the other side, the safety coefficient and hazards of mining products have been improved.
- 5) The presented framework compares the different scenarios and the competitive state-of-the-art experiments are fully evaluated. The experiment’s results demonstrate and verify the efficacy and performance. To the best of our knowledge, our presented ConFrame framework is the newest cutting-edge in the mining industry worldwide, it has the value of extended for other processes industry.

The remainder of the article is organized as follows. Section II gives some background and problem formulation. Section III introduces the process of the proposed framework. The industrial cases experiments are included in Section IV. Finally, the concluding remarks and future considerations are given in Section V.

II. PROBLEM DESCRIPTION AND PIPELINE FORMULATION

The conventional mining process has the obstacle of missing a standard intelligent platform, which generates challenges such as low efficiency, high labor cost, and high dangers. The

TABLE I
COMPARISON SUMMARY OF SOME STATE-OF-THE-ART WORKS FOR INTELLIGENT METHODS

Symbol	Convergence	Optimization	Robustness	DCS Application	Intelligence	Stability
Zhang <i>et al.</i> [22]	Not feasible	Local	Feasible	Not feasible	Feasible	Feasible
Li <i>et al.</i> [26]	Not feasible	Local	Feasible	Feasible	Not Feasible	Feasible
Ma <i>et al.</i> [37]	Feasible	Global	Not feasible	Feasible	Feasible	Not feasible
Lei <i>et al.</i> [28]	Not feasible	Local	Feasible	Not feasible	Not Feasible	Feasible
Sun <i>et al.</i> [17]	Not feasible	Global	Feasible	Not feasible	Feasible	Not feasible
Wang <i>et al.</i> [25]	Not feasible	Local	Feasible	Feasible	Not feasible	Feasible
Schent <i>et al.</i> [26]	Feasible	Local	Not feasible	Not feasible	Feasible	Feasible
Kao <i>et al.</i> [18]	Not feasible	Local	Feasible	Not feasible	Feasible	Feasible

underflow concentration of the deep cone thickener is the key index and is critical to the effectiveness of the paste-filling mining process. Specifically, accidents like obstruction and crushing may occur if the concentration is too high. If the concentration is too low, the subterranean filling body's intensity will be insufficient, posing a safety risk. To accomplish safe and effective paste-filling mining, it is required to understand the changing rules of the deep cone thickener's underflow concentration and precisely estimate the underflow concentration. The forecast of the deep cone thickener's underflow concentration serves as the foundation for employees to carry out paste-filling mining control. The existing industry apparently lacks such kind of framework with high performance. A novel intelligent platform is motivated and developed to address the difficulty of the existing pasting-filling process.

The requirement and farsightedness of mining is the smarting mining and intelligent development. However, the challenge that existed in the industry is the lack of smart technologies applied to build the "One-key-work" system. The main procedures of the paste-filling process are given in Fig. 1. The main reaction in the plant-wide of mining paste-filling process is that the low concentration tail particle (20%–30%) are fed to the deep cone system, under the function of the gravity of DCT, the high stable underflow concentration of tail and then be transferred to the pump. The flocculant work as the catalyst to accelerate the paste precipitation process. The other relevant process key-quality variables existing in the DCT are specified by ①–⑩. A stable underflow concentration of paste is then formalized at the bottom of the DCT. The final process of the pasting filling is finished by the pump.

The stable underflow concentration is then pumped to the underground mining to paste the dependency. The whole process is time-consistence and with multiple variable reaction mechanisms.

With the development of the distributed control system, the multi-interaction of the computers and the control are more feasible, and a higher efficacy is achieved.

A parenthetical statement of the designed DCS for the underflow paste filling system is given in Fig. 2. The prediction framework is based on the stacked bidirectional long-short time memory (BiLSTM) (see Algorithm 1). Then the prediction method is further used as the control objective function for the whole pasting mining optimization process (see Algorithm 2). Finally, the prediction preparation phase model and the optimal control output jointly formulate the ConFrame idea implementation and deployment (see Algorithm 3). We then

Algorithm 1 BiLSTM-Based Prediction Preparation Phase Implementation (Stage 1)

Require:

- The input variable set;
- The set of prediction network architectures;
- The batch size, learning rate, moving sliding window size, and prediction steps;

Ensure:

- The prediction model f_{pre} , the weights of the BiLSTM;
- 1: Collect the sufficient samples with the input variables matrix $[C_i, Q_i, F, C_o, Q_o]$;
- 2: Initialize the network parameters which include the time sliding window T , prediction steps s , the input time sequence L , the layers of forward and reverse BiLSTM, and the average filter step matrix;
- 3: Training the whole network according to the designed equations (8)–(29), all the gradients of the partial derivative of the error can be computed and the weights are tuned accordingly;
- 4: Build a training window with the duration T , slide the data loaded for neural network training, and the loss function is the root mean square error between the predicted value and the true value;
- 5: Once the training is finished, the end-to-end underflow concentration prediction model in industrial CTS is obtained, which can recursively predict the underflow concentration at the next T moment according to the feed concentration, feed flow rate, underflow flow rate, thickener pressure, and underflow concentration at the previous time of T duration.
- 6: **return** f_{pre} ;

will move the focus to the implementation of the prediction stage (see Fig. 3).

A. Stage 1: Prediction Approach

As given all the details of the prediction and control, then a novel intelligent pasting filling platform is then developed. The LSTM algorithm with filtering technique is utilized as the initialization of the optimal framework. *Given*: the sequential and historical of quality variables series $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n, \mathbf{Y})$, our motivation is to: *Predict*: the next T instant of the underflow concentration $\hat{\mathbf{Y}} = (y_{T+1}, y_{T+2}, \dots, y_{T+T})^T \in \mathbb{R}^T$. Specifically,

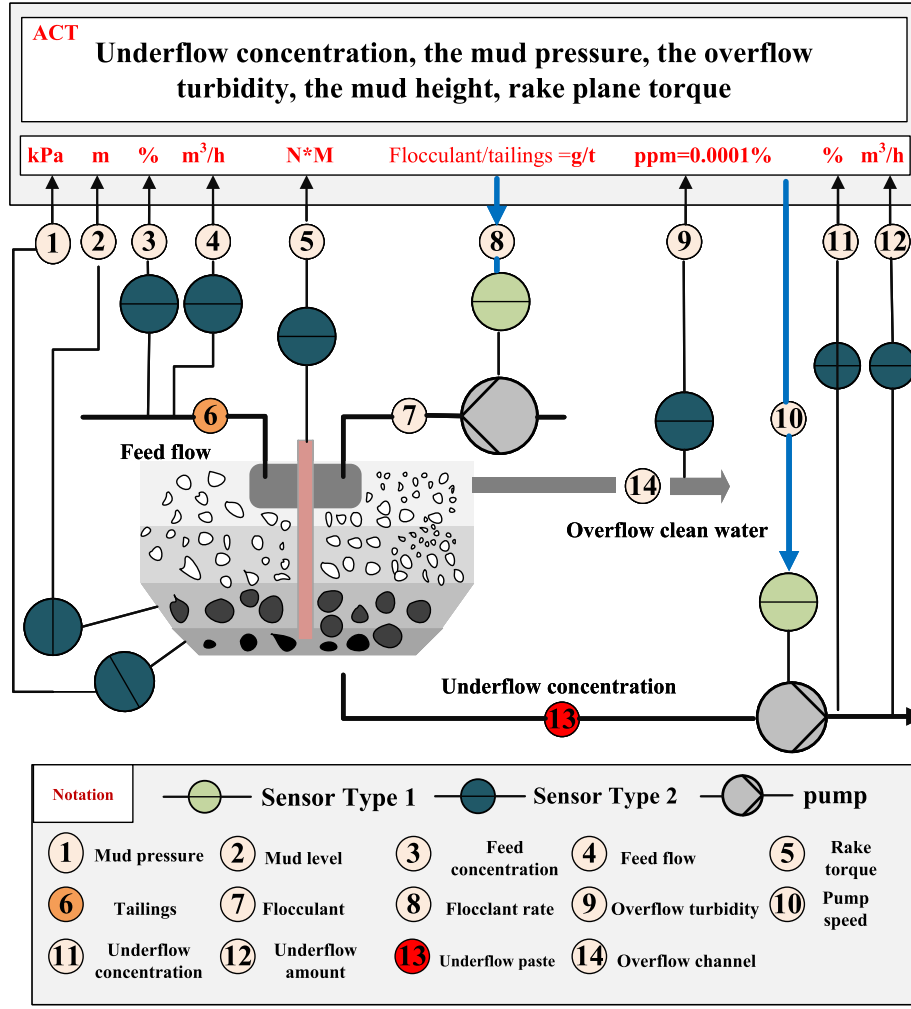


Fig. 1. Test bench of deep cone thicker treatment plant process and the role of our proposed method. The blue circle represents the sensor type 1 and the blue symbolizes type 2, respectively. The gray circle represents the pump unit. All the main process key-quality variables (①-⑭) have been specified.

we need to learn a nonlinear mapping representation with $(\hat{y}_{T+1}, \hat{y}_{T+2}, \dots, \hat{y}_{T+\tau}) = F(y_1, \dots, y_T, x_1, \dots, x_n)$.

The next stage is to train the whole network parameters. Consider the dual Bi-LSTM has two embeddings with feed-forward LSTM and the reverse LSTM. Thus, the AF-BiLSTM fine-tuning policy for the hyper-parameters is designed as the following equations.

Suppose the feedforward and reverse input weights: $\vec{W}_z, \vec{W}_i, \vec{W}_f, \vec{W}_o, \vec{W}_z, \vec{W}_i, \vec{W}_f, \vec{W}_o \in \mathbb{R}^{N \times M}$.

Recurrent feedforward and reverse weights: $\vec{R}_z, \vec{R}_i, \vec{R}_f, \vec{R}_o, \vec{R}_z, \vec{R}_i, \vec{R}_f, \vec{R}_o \in \mathbb{R}^{N \times N}$.

Output feedforward and reverse weights: $\vec{p}_i, \vec{p}_f, \vec{p}_o, \vec{p}_i, \vec{p}_f, \vec{p}_o \in \mathbb{R}^N$.

Bias weights: $\vec{b}_z, \vec{b}_i, \vec{b}_f, \vec{b}_o, \vec{b}_z, \vec{b}_i, \vec{b}_f, \vec{b}_o \in \mathbb{R}^N$.

The piecewise epoch equations would be

$$\vec{z}_f^t = \vec{W}_z \mathbf{x}_f^t + \vec{R}_z \mathbf{y}_f^{t-1} + \vec{b}_z \quad (1)$$

$$\mathbf{z}_f^t = g(\vec{z}_f^t). \quad (2)$$

The bidirectional forward and reverse input gates are

$$\bar{\mathbf{i}}_f^t = \vec{W}_i \mathbf{x}_f^t + \vec{R}_i \mathbf{y}_f^{t-1} + \vec{p}_i \odot \vec{c}^{t-1} + \vec{b}_i \quad (3)$$

$$\mathbf{i}_f^t = \sigma(\bar{\mathbf{i}}_f^t) \quad (4)$$

$$\bar{\mathbf{i}}_r^t = \vec{W}_i \mathbf{x}_r^t + \vec{R}_i \mathbf{y}_r^{t-1} + \vec{p}_i \odot \vec{c}^{t-1} + \vec{b}_i \quad (5)$$

$$\mathbf{i}_r^t = \sigma(\bar{\mathbf{i}}_r^t). \quad (6)$$

The forward and reverse forget gates are represented as

$$\bar{\mathbf{f}}_f^t = \vec{W}_f \mathbf{x}_f^t + \vec{R}_f \mathbf{y}_f^{t-1} + \vec{p}_f \odot \vec{c}^{t-1} + \vec{b}_f \quad (7)$$

$$\mathbf{f}_f^t = \sigma(\bar{\mathbf{f}}_f^t) \quad (8)$$

$$\bar{\mathbf{f}}_r^t = \vec{W}_f \mathbf{x}_r^t + \vec{R}_f \mathbf{y}_r^{t-1} + \vec{p}_f \odot \vec{c}^{t-1} + \vec{b}_f \quad (9)$$

$$\mathbf{f}_r^t = \sigma(\bar{\mathbf{f}}_r^t). \quad (10)$$

The corresponding cell states are calculated by

$$\mathbf{c}_f^t = \mathbf{z}_f^t \odot \mathbf{i}_f^t + \mathbf{c}_f^{t-1} \odot \mathbf{f}_f^t \quad (11)$$

$$\mathbf{c}_r^t = \mathbf{z}_r^t \odot \mathbf{i}_r^t + \mathbf{c}_r^{t-1} \odot \mathbf{f}_r^t. \quad (12)$$

Finally, the output of dual bidirectional LSTM is

$$\bar{\mathbf{o}}_f^t = \vec{W}_o \mathbf{x}_f^t + \vec{R}_o \mathbf{y}_f^{t-1} + \vec{p}_o \odot \mathbf{c}_f^t + \vec{b}_o \quad (13)$$

$$\mathbf{o}_f^t = \sigma(\bar{\mathbf{o}}_f^t) \quad (14)$$

$$\bar{\mathbf{o}}_r^t = \vec{W}_o \mathbf{x}_r^t + \vec{R}_o \mathbf{y}_r^{t-1} + \vec{p}_o \odot \mathbf{c}_r^t + \vec{b}_o \quad (15)$$

Algorithm 2 Improved PSO-Based Optimal Control Model Implementation (Stage 2)**Require:**

The prediction network model f_{pre} ;
 The optimal control model with constraints
 Iteration epochs k , error bound η ;

Ensure:

The global optimization control parameters, fitness value f_i and global fitness value f_{id} ;

- 1: Initialize N_{pop} , maximum generation gen_{max} , position X_{id} , and velocity V_{id} ;
- 2: **for** $t=1:gen_{max}$
- 3: **for** $i=1:pop_{size}$
- 4: run the prediction model f_{pre} and evaluate the fitness of the particles f_i
- 5: find the P_{id} is the best previous position of the i th particle; P_{gd} is the best previous position among all particles in the swarm
- 6: **if** $f_i < f_{gd}$ **then break**
- 7: **else if** $f_i > f_{gd}$ **then** update the prediction network inertia weight W
- 8: **end**
- 9: update the velocity and position
- 10: check the maximum number of iterations
- 11: **if** $x_{i,d}(t+1) < \eta$ **then** back to step 4;
- 12: **else break**
- 13: **end**
- 14: **end**
- 15: **return** optimal control Q_1, C_1, k, Q_2 ;

Algorithm 3 ConFrame Framework Implementation**Require:**

The maximum iteration step;
 The trained prediction model, f_{pre} ;
 Ensemble of prediction network on control method, U_{n-1} ;

Ensure:

Ensemble of classifiers on the global optimization parameters, E_n and fitness value η ;

- 1: Operate prediction stage with **Algorithm 1**;
- 2: Operate optimal control stage with **Algorithm 2**;
- 3: Update the weight parameters in f_{pre} , $k=k+1$, back step 1;
- 4: **return** Optimal control input for the PLC execution unit;

$$\mathbf{o}_r^t = \sigma(\bar{\mathbf{o}}_r^t) \quad (16)$$

$$h_i^t = h(\mathbf{c}_f^t) \odot \mathbf{o}_f^t \quad (17)$$

$$(h')_i^t = h(\mathbf{c}_r^t) \odot \mathbf{o}_r^t \quad (18)$$

where σ , g , and h are the piecewise nonlinear activation projections, respectively. Since the backward and forward processes are the same in principle but in opposite series order, the equation of the backward process could be derived similarly by replacing $\rightarrow$ with $\leftarrow$. Then we finished the encoder representation of the underflow key quality features, the attention mechanism inherited from the BiLSTM output

can be given as the following embedding feedforward process:

$$\text{soft}_i = \tanh(\vec{W}_h h_i + \vec{b}_h) + \tanh(\vec{W}_h h'_i + \vec{b}_h) \quad (19)$$

$$\alpha_i = \frac{\exp(\text{soft}_i)}{\sum_{i=1}^N \exp(\text{soft}_i)}, \quad \sum_{i=1}^N \alpha_i = 1 \quad (20)$$

$$y = \sum_{i=1}^N \alpha_i h_i, \quad y' = \sum_{i=1}^N \alpha_i h'_i. \quad (21)$$

The dual-attention long-short time memory is developed from [13], which is rewritten as the following expression:

$$p_j = \sum_{j=1}^s W_j \cdot y_i(1, \dots, L) = \sum_{j=1}^s \sum_{i=1}^L W_j W_y \cdot \langle y_i, y'_i \rangle \quad (22)$$

where $W_j \in \mathbb{R}^{N \times L}$, $W_y \in \mathbb{R}^{N \times S}$ are the weight parameter that connects the dense layer and attention layer, respectively. Thus, the aim for the whole of stage 1 is to learn a mapping prediction function

$$\begin{aligned} p(y_{t+s}|X_1, X_2, \dots, X_L, X_{L+k}, X_{L+k+1}, \dots, X_{2L+k-1}) \\ = p(y_{t+s}|\langle C_i(t), Q_i(t), F(t), C_o(t), Q_o(t) \rangle) \\ = p\left(y|W \cdot \sum_{i=1}^L \langle y_i, y'_i \rangle\right) \end{aligned} \quad (23)$$

to this end, the prediction network for Conframe is finished.

B. Global Approximation Ability Analysis for Conframe

The global approximation ability theory is given in this part. Before introducing the main theorem, a lemma is necessary at this point.

Lemma 1: For any piecewise continuous function $F(x)$ on a compact space set D , the stacked network with enough neurons can appropriate any nonlinear mapping function precisely, and the approximate error is bounded by an arbitrarily small error interval $\varepsilon \in \mathbb{R}^+$ on D .

Theorem 1: Consider the nonlinear discrete-time system, the proposed Conframe algorithm. Let the momentum-based back-propagation through time (BPTT) update the weights of the prediction network. The proper iterative PSO method is used to obtain the specified objective function. With the increase of the number of hidden layers equals to infinite, then, there exists positive energy Lyapunov function $\mathbf{E}_L = (1/2) \sum_{i=1}^N \|e_i\|^2$ and exists the positive small error constant ε , ς , and ξ that makes the residual error limited to an arbitrarily small positive boundary $\varepsilon + \varsigma + 2\xi$, and also meet the uniformly ultimately bounded condition

$$\nabla \leq \bar{\nabla} \leq \varepsilon + \varsigma + 2\xi. \quad (24)$$

Proof: Consider the iterative error of the prediction network from L to $L+1$, with the output from dual-attention LSTM in (1), the above (17) can be represented as

Define $\sum_{i=1}^L \langle y_i, y'_i \rangle = \phi_L$, $\sum_{i=1}^{L+1} \langle y_i, y'_i \rangle = \phi_{L+1}$, $\nabla' = \nabla - \|y_{L+1} - y_L\|^2$, then the following equation is given:

$$\begin{aligned} \nabla' = & \left\| \sigma \left(W_s W_y \cdot \phi_{L+1} \cdot \sum_{j=1}^s W_j \right) \right\|^2 \\ & - 2 \left\langle y_{L+1}, \sigma \left(W_s W_y \cdot \phi_{L+1} \cdot \sum_{j=1}^s W_j \right) \right\rangle \end{aligned}$$

$$+ 2 \left\langle y_L, \sigma \left(W_s W_y \cdot \phi_L \cdot \sum_{j=1}^s W_j \right) \right\rangle - \left\| \sigma \left(W_s W_y \cdot \phi_L \cdot \sum_{j=1}^s W_j \right) \right\|^2. \quad (27)$$

According to Lemma 1, with the increase of the L layer, the square residual error would converge an error bound constant $\|\sigma(W_s W_y \cdot \phi_{L+1} \cdot \sum_{j=1}^s W_j)\|^2 - \|\sigma(W_s W_y \cdot \phi_L \cdot \sum_{j=1}^s W_j)\|^2 < \varsigma + O_D^2(\phi_{L+1}, \phi_L)$. Similarly, $\langle y_L, \sigma(W_s W_y \cdot \phi_L \cdot \sum_{j=1}^s W_j) \rangle - \langle y_{L+1}, \sigma(W_s W_y \cdot \phi_{L+1} \cdot \sum_{j=1}^s W_j) \rangle \leq \xi + O_D(\phi_{L+1}, \phi_L)$. According to the quality of Cauchy-Bunyakovsky-Schwarz inequality, then the above equation is

$$\begin{aligned} \nabla' &\leq \bar{\nabla}' = \varsigma + 2\xi + O_D^2(\phi_{L+1}, \phi_L) + 2O_D(\phi_{L+1}, \phi_L) \\ &\propto \varsigma + 2\xi + O_D^2(\phi_{L+1}, \phi_L) \\ &\doteq \varsigma + 2\xi \end{aligned} \quad (28)$$

therefore, the first term in equation (18) can approximate an error upper boundary, $\|y_{L+1} - y_L\|^2 \leq \varepsilon + O_D^2(e(y_{L+1}, y_L))$, where $e(y_{L+1}, y_L)$ is the similarity distance measure between two different samples y_{L+1} and y_L . Consequentially, equation (17) can finally be rewritten as

$$\begin{aligned} \nabla &= \|y_{L+1} - y_L\|^2 + \nabla' \\ &\leq \varepsilon + O_D^2(e(y_{L+1}, y_L)) + \varsigma + O_D^2(\phi_{L+1}, \phi_L) \\ &\quad + \xi + O_D(\phi_{L+1}, \phi_L) \\ &= \varepsilon + \varsigma + 2\xi. \end{aligned} \quad (29)$$

Then the global approximation ability proof is complete.

C. Stage 2: Optimal Control Approach

1) *Objective Function*: The first goal is to consider the main control parameter of the deep cone thickener of bottom flow concentration, the main control effect of the model is to maintain the bottom flow concentration at the best setting value and to be able to regulate the deep cone thickener by setting the parameters in advance so that the underflow concentration of deep cone thickener can be maintained to fluctuate within the setting value and to meet the smallest possible residual error to achieve the best control accuracy.

Underflow concentration C_2 is denoted as our main control target, according to the actual production of the mine fill demand, needs to meet in the set controllable range as large as possible, here the prediction model as a function of the output adaptation function, combined into the prediction algorithm objective function can be expressed as follows:

$$\text{Max: } J = f_{\text{Bi-LSTM}}(C_1, Q_1, H, k, Q_2, C_2, \Delta t) \quad (30)$$

where the fitness function J can be used as the objective function of bottom flow concentration in the model, and f_{LSTM} is the bottom flow concentration prediction algorithm, $C_1, Q_1, H, k, Q_2, C_2, \Delta t$ denoting: feed concentration, feed flow rate, mud layer height, elasticity coefficient, discharge flow rate, bottom flow concentration, and reaction delay time, respectively. Considering the correlation relationship between mud layer pressure and mud layer height, the mud layer height can be selected for prediction, and the mud layer pressure is

$$\begin{aligned} \nabla &= \|e_{L+1}\|^2 - \|e_L\|^2 \\ &= \left\| y_{L+1} - p_{L+1} \left(y_{L+1} | W_{L+1} \sum_{i=1}^{L+1} \langle y_{i+1}, y'_{i+1} \rangle \right) \right\|^2 - \left\| y_L - p_L \left(y_L | W_L \sum_{i=1}^L \langle y_i, y'_i \rangle \right) \right\|^2 \\ &= \|y_{L+1} - y_L\|^2 + \left\| p_{L+1} \left(y_{L+1} | W_{L+1} \sum_{i=1}^{L+1} \langle y_{i+1}, y'_{i+1} \rangle \right) \right\|^2 - 2 \left\langle y_{L+1}, p_{L+1} \left(y_{L+1} | W_{L+1} \sum_{i=1}^{L+1} \langle y_{i+1}, y'_{i+1} \rangle \right) \right\rangle \\ &\quad + 2 \left\langle y_L, p_L \left(y_L | W_L \sum_{i=1}^L \langle y_i, y'_i \rangle \right) \right\rangle - \left\| p_L \left(y_L | W_L \sum_{i=1}^L \langle y_i, y'_i \rangle \right) \right\|^2 \end{aligned} \quad (25)$$

$$\begin{aligned} \nabla &= \|y_{L+1} - y_L\|^2 + \|\sigma(W_s \cdot p_{L+1})\|^2 - 2\langle y_{L+1}, \sigma(W_s \cdot p_{L+1}) \rangle + 2\langle y_L, \sigma(W_s \cdot p_{L+1}) \rangle - \|\sigma(W_s \cdot p_L)\|^2 \\ &= \|y_{L+1} - y_L\|^2 + \left\| \sigma \left(W_s \cdot \sum_{j=1}^s W_j \cdot y_{i(i=2, \dots, L, L+1)} \right) \right\|^2 - 2 \left\langle y_{L+1}, \sigma \left(W_s \cdot \sum_{j=1}^s W_j \cdot y_{i(i=2, \dots, L, L+1)} \right) \right\rangle \\ &\quad + 2 \left\langle y_L, \sigma \left(W_s \cdot \sum_{j=1}^s W_j \cdot y_{i(i=1, \dots, L)} \right) \right\rangle - \left\| \sigma \left(W_s \cdot \sum_{j=1}^s W_j \cdot y_{i(i=1, \dots, L)} \right) \right\|^2 \\ &= \|y_{L+1} - y_L\|^2 + \left\| \sigma \left(W_s \cdot \sum_{j=1}^s \sum_{i=2}^{L+1} W_j W_y \cdot \langle y_{i+1}, y'_{i+1} \rangle \right) \right\|^2 - 2 \left\langle y_{L+1}, \sigma \left(W_s \cdot \sum_{j=1}^s \sum_{i=2}^{L+1} W_j W_y \cdot \langle y_{i+1}, y'_{i+1} \rangle \right) \right\rangle \\ &\quad + 2 \left\langle y_L, \sigma \left(W_s \cdot \sum_{j=1}^s \sum_{i=1}^L W_j W_y \cdot \langle y_i, y'_i \rangle \right) \right\rangle - \left\| \sigma \left(W_s \cdot \sum_{j=1}^s \sum_{i=1}^L W_j W_y \cdot \langle y_i, y'_i \rangle \right) \right\|^2 \end{aligned} \quad (26)$$

used as the constraint of the variable, thus simplifying the optimization model.

The paste-filling production requires an actual human setting to track the underflow concentration, and thus the objective function needs to be adjusted, on the basis of the above equation, this article further proposes an improved objective function, taking into account the human setting, time delay and other factors, and constructs the following optimization objective:

$$\text{Min: } J' = C_{\text{set}} - f_{\text{Bi-LSTM}}(C_1, Q_1, H, k, Q_2, C_2, \Delta t). \quad (31)$$

C_{set} is the set value of underflow concentration. To obtain the most precise tracking possible, the maximum value of the goal function's adaption of the underflow concentration is modified to the lowest value of the objective function's set value and the inaccuracy of the anticipated value. The goal function may best meet and match the mine's actual production demands, achieving a very excellent tracking setting need, which is the main enhancement to realize the intelligent model and combine it with industrial practice to apply.

2) *Constraints*: The creation of a good and accurate objective function is also the priority in the realization of the model approach, which must be combined with the production process and mechanism, data knowledge to consider the constraints of the model variables, combined with the optimization of the objective function, in order to solve the intelligent control of the deep cone thickener production process.

Based on the adaptive "Determined Equivalence Principle," "Separate Design Principle," and "Material Balance Principle," the controller is designed to meet the provided performance index. The matching controller is built using the specified performance metric. The parameter values of the controlled object are then approximated online using parameter recognition, and the system is controlled. The first consideration in the deep cone thickener manufacturing process is material balance. Throughout the manufacturing process, the material does not strictly meet the following mathematical requirements:

$$C_1 Q_1 = C_2 Q_2. \quad (32)$$

The main reason is based on its production parameters for sensor monitoring parameters, such as concentration meter, flowmeter, etc. There is a certain range of accuracy and measurement error, resulting in the actual database of statistical variables there are errors; In addition, the production process will be affected by the pipe diameter, sedimentation tank adhesion, settlement losses, and other factors, in order to compensate for the progress of the material balance, this article introduces the elasticity coefficient k to adjust the corresponding model so that it meets the following equation constraints. We then rewrite the above equation as

$$C_1 Q_1 = k C_2 Q_2 \quad (33)$$

where $0 < k \leq k^{\text{max}}$. The correlation link between variables is evaluated, and the elasticity coefficient k is adjusted to fine-tune the model to make it dynamically adapt to the conservation rule and boost the model's resilience. Meanwhile, the mud layer height, mud layer pressure, feed concentration, feed flow rate, bottom flow rate, and bottom flow concentration must all fall within specific ranges, and the mud layer heights

H_{min} and H_{max} must be set to fulfill

$$H^{\text{min}} \leq H \leq H^{\text{max}}. \quad (34)$$

Feed concentration C_1 , feed flow rate Q_1 , discharge flow rate Q_2 , underflow concentration C_2 , the mud layer pressure P satisfy the following constraints, respectively:

$$C_1^{\text{min}} \leq C_1 \leq C_1^{\text{max}} \quad (35)$$

$$Q_1^{\text{min}} \leq Q_1 \leq Q_1^{\text{max}} \quad (36)$$

$$Q_2^{\text{min}} \leq Q_2 \leq Q_2^{\text{max}} \quad (37)$$

$$C_2^{\text{min}} \leq C_2 \leq C_2^{\text{max}} \quad (38)$$

$$P^{\text{min}} \leq P \leq P^{\text{max}}. \quad (39)$$

Considering the relationship between the mud layer height and the mud layer pressure, there are the following constraints:

$$P = s * H. \quad (40)$$

In summary, the optimization conditions satisfied by this optimization model are as follows:

$$\text{s.t.} \begin{cases} C_1 Q_1 = k C_2 Q_2 \\ H = s * P \\ H^{\text{min}} \leq H \leq H^{\text{max}} \\ 0 \leq k \leq k^{\text{max}} \\ C_1^{\text{min}} \leq C_1 \leq C_1^{\text{max}} \\ Q_1^{\text{min}} \leq Q_1 \leq Q_1^{\text{max}} \\ Q_2^{\text{min}} \leq Q_2 \leq Q_2^{\text{max}} \\ C_2^{\text{min}} \leq C_2 \leq C_2^{\text{max}} \\ P^{\text{min}} \leq P \leq P^{\text{max}}. \end{cases} \quad (41)$$

Overall, the optimization problem of the DCS can be formulated as follows:

$$\text{Objective: } \min J' = C_{\text{set}} - f_{\text{pre}}(C_1, Q_1, H, k, Q_2, C_2, \Delta t)$$

$$\text{s.t.} \begin{cases} C_1 Q_1 = k C_2 Q_2 \\ H = s * P \\ H^{\text{min}} \leq H \leq H^{\text{max}} \\ 0 \leq k \leq k^{\text{max}} \\ C_1^{\text{min}} \leq C_1 \leq C_1^{\text{max}} \\ Q_1^{\text{min}} \leq Q_1 \leq Q_1^{\text{max}} \\ Q_2^{\text{min}} \leq Q_2 \leq Q_2^{\text{max}} \\ C_2^{\text{min}} \leq C_2 \leq C_2^{\text{max}} \\ P^{\text{min}} \leq P \leq P^{\text{max}} \end{cases} \quad (42)$$

where Q_1 , C_1 , k , and Q_2 are the control variables which need to be optimized. The above optimal problem is a nonlinear mixed integer with uncertainty parameters programming problem. The control demand is to minimize the objective function by dynamically controlling the feed flow Q_1 and underflow flow Q_2 based on the current elasticity coefficient k . As a result, our focus in solving (42) is on locating the optimal control parameters (specifically Q_1 and Q_2) in the feasible solution space. During the deployment process, the optimal output would be directly transferred to the programmable logic controller (PLC) unit to implement the optimal control demand. Due to the convex feature of the original problem, the dual problem can be further given. The conventional optimization solution cannot obtain a closed solution. Therefore, we introduce an improved PSO to implement this problem.

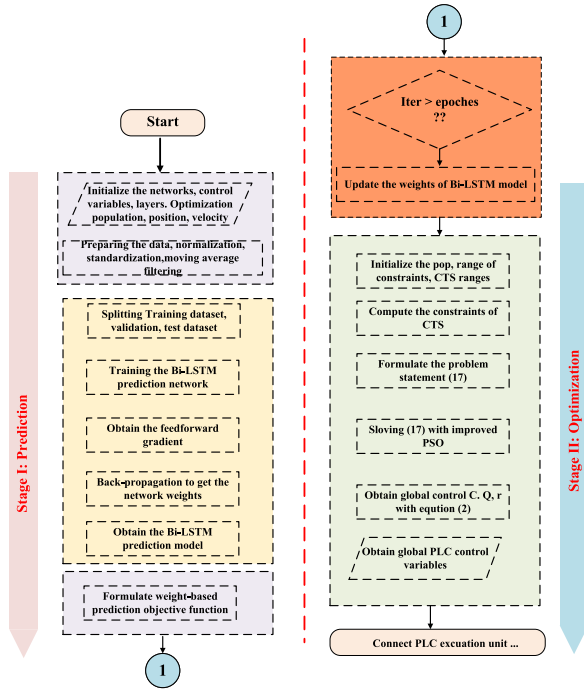


Fig. 2. Illustrative flowchart of the proposed algorithm for CTS.

D. Improved PSO-Based Optimization Process

The underflow concentration optimization problem in cone thickener system (CTS) has been clearly presented in the above description. The next step is to solve the proposed control and find a global solution for the control successor

$$\begin{aligned} V_{id}(t) &= w(t) \times V_{id}(t-1) + c_1 \times \text{Rand}() \times (P_{id} - X_{id}) \\ &\quad + c_2 \times \text{Rand}() \times (P_{gd} - X_{id}) \\ X_{id}(t) &= X_{id}(t-1) + V_{id}(t) \end{aligned} \quad (43)$$

where V_{id} is the current velocity of i th particle $i = 1, \dots, n$, where n is the population size; $\text{Rand}()$ represents a uniform random number between 0 and 1; subscript d is the dimension of the particle; P_{id} is the best previous position of the i th particle; P_{gd} is the best previous position among all particles in the swarm; and X_{id} is the current position of the i th particle. The strategy for this issue is the optimization swarm optimization algorithm (see Algorithm 2).

Therefore, the core algorithm brain procedure is given in flowchart 2.

III. PROPOSED CONFRAME PLATFORM FORMULATION

In this section, the ConFrame intelligent platform is given. There is a full-scale process framework for the mining industry control. The basic first stage is the prediction task based on the BiLSTM with smooth time filtering. The total full-scale framework used in an illustrative pipeline is depicted in Fig. 3.

A. Convergence Analysis

In this section, the convergence of the proposed algorithm will be presented, in which the iterative cost function $J_{i,j}$ and the iterative control law u_i converge to the optimal solution J^* and u^* , respectively. Our next goal is that with the time

instant j tends to infinity, the cost function sequence is a monotonous increasing, and $J_j(u, \phi, k)$ will converge to a bounded upper function $\bar{J}_j(u, \phi, k)$ as $j \rightarrow \infty$. Before giving our theorem, we introduce two lemmas from [24], some extended convergence analysis can be seen in [23].

Lemma 2: For all $w, \phi_1 = c_1 r_1$ and $r_1 \phi_2 = c_2 r_2$ falling inside the convergent parameter domain, then exists a finite N and a positive ε , for all $k \geq N$

$$\|J^k(x_{i,t}) - J^{k+1}(x_{i,t})\| < \varepsilon. \quad (44)$$

Lemma 3: Under the constraints of defined subset convergent region of $\delta(u) \rightarrow \delta(u^*)$, all the particles $1 \leq i \leq h$ will converge onto the global position $J^*(x_{i,t})$, so that

$$\lim_{t \rightarrow +\infty, u \rightarrow u^*} (x_{i,t}) = J^*(x_{i,t}). \quad (45)$$

Theorem 2: Let's $J_j(u, \phi, k)$, u_i be achieved by (9)–(21), and the iterative cost function $J_j(u, \phi, k)$ is a monotonically nonincreasing function for $\forall j \geq 0$

$$\|J_{j+1}(u, \phi, k) - J_j(u, \phi, k)\| \leq \varepsilon \quad (46)$$

and then, the iterative cost function converges to the optimal cost function $J^*_{j+1}(u, \phi, k)$ as $j \rightarrow \infty$

$$\lim_{j \rightarrow \infty} J_j(u, \phi, k) = J^*_{j+1}(u, \phi, k). \quad (47)$$

Proof: Let $J_j(u, \phi, k) = J_{j,\infty}(u, \phi, k)$. For the initial state $j = 0$, $J_{0,\infty}(u, \phi, k) = \tilde{J}_{0,\infty}(u, \phi, k)$, $u_0(x_t) = \tilde{u}_0(x_t)$. The goal is to obtain the solution of

$$\tilde{u}(x_t) = \arg \min_u J(u, \phi, k) \quad (48)$$

according to Lemma 2, let $J_{j,\infty}(u, \phi, k) = \tilde{J}_{j,\infty}(u, \phi, k)$, $u_j(x_t) = \tilde{u}_j(x_t)$, then

$$\begin{aligned} J_{j,\infty}^{k+1}(u, \phi, k) &= \lim_{j \rightarrow \infty} \tilde{J}_{j,\infty}^{k+1}(u, \phi, k) \leq \tilde{J}_{j,1}^{k+1}(u, \phi, k) \\ &= U^k(x_{i,t}, \phi, k) + \tilde{J}_{j,1}^k(u, \phi, k) \\ &\leq U^{k-1}(x_{i,t}, \phi, k) + \tilde{J}_{j,1}^{k-1}(u, \phi, k) \\ &= J_{j,\infty}^k(u, \phi, k) = \tilde{J}_{j,\infty}^k(u, \phi, k) \end{aligned} \quad (49)$$

where $U^{k-1}(x_{i,t}, \phi, k)$ is the optimal control space set. Using the positive ε to approximate the error between $J_{j,\infty}^{k+1}(u, \phi, k)$ and $J_{j,\infty}^k(u, \phi, k)$, the following equation is represented:

$$\|J_{j+1}(u, \phi, k) - J_j(u, \phi, k)\| \leq \varepsilon \quad (50)$$

therefore, (46) is obtained.

According to Lemma 3, due to the reason that the $J_j(u, \phi, k) > 0$ is non-increasing sequences that satisfy the loss improvement with sequences. Hence, $\lim_{j \rightarrow \infty} J_j(u, \phi, k)$ can be replaced by the expression $J_\infty(u, \phi, k)$, and $u(x_t, \phi, k) = \arg \min_u J_\infty(u, \phi, k)$, substituting $J_\infty(u, \phi, k)$ into (45), obtains

$$\begin{aligned} \lim_{j \rightarrow \infty} J_j(u, \phi, k) &= J_\infty(u, \phi, k) \\ &= U^k(x_{i,t}, \phi, k) + \tilde{J}_{j,1}^k(u, \phi, k) \\ &= J^*_{j+1}(u, \phi, k). \end{aligned} \quad (51)$$

Then this proof is complete.

B. Stability Analysis for ConFrame

The stability of the proposed ConFrame is analyzed in this section to find the regions of asymptotic stability. According

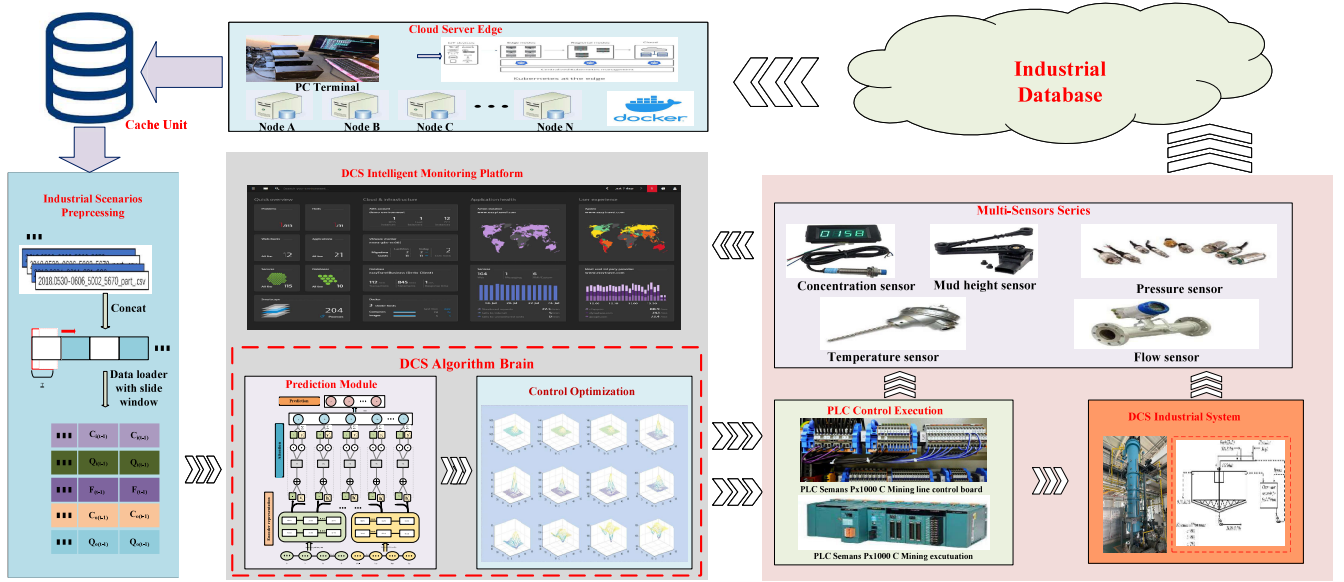


Fig. 3. Deployment working pipeline of the proposed intelligent platform for the deep CTS.

to the previous Lemma 1–3, and Theorem 2, the following theorem is concluded.

Theorem 3: Consider the prediction model f_{pre} , the optimal function $J^*(x_{i,t})$ with the constraints. Let the momentum-based gradient descent method be used to update the weights of the time-sequence prediction network. The optimal solution is feasible in the compact subspace $\delta(u)$. Then, there exist positive constants α_Λ , satisfying

$$0 \leq \alpha_\Lambda \leq \frac{1}{\|\varepsilon\|^2} \quad (52)$$

at the same time, and the following conditions are also met:

$$\begin{aligned} b_{\tilde{W}_\Lambda} &\equiv \sqrt{\frac{J_2(u)}{1 - \alpha_\Lambda \|\varepsilon\|^2}} < \|\xi_{\Lambda,k}\| \\ b_{\tilde{W}_c} &\equiv \sqrt{\frac{J_2(u)}{1 - 4\alpha_c \|\varepsilon\|^2 + \alpha_c \|\varepsilon_{\text{fpre}}\|^2}} < \|\xi_{c,k}\|. \end{aligned} \quad (53)$$

Following this, the final output estimation error $\tilde{W}_\Lambda$ of ConFrame is UUB.

Proof: Consider the Lyapunov function candidate

$$L_k = L_{\tilde{W}_{\Lambda,k}} \quad (54)$$

where $L_{\tilde{W}_{\Lambda,k}} = (1/\alpha_\Lambda) \text{tr}(\tilde{W}_\Lambda^T(k) \tilde{W}_\Lambda(k))$. Here, $\text{tr}(\cdot)$ denotes the trace operator. It is worth noting that, compared with j that should be presented with i simultaneously, k denotes the iteration index.

Taking the first discrete difference of (54), an equation can be obtained

$$\Delta L_k = \Delta L_{\tilde{W}_{\Lambda,k}}. \quad (55)$$

In the following, the terms in (55) will be analyzed separately. First, $\Delta L_{\tilde{W}_{\Lambda,k}}$ will be considered.

According to Lemma 2, the optimal prediction model function can be represented as

$$\Lambda^*(u) = \sum_{j=1}^s \sum_{i=1}^L W_j^* \cdot \langle y_i, y_i' \rangle \sigma(u) + \varepsilon_\Lambda(k) \quad (56)$$

where W_Λ^* denotes the optimal weights of the prediction BiLSTM network and its upper bound is $\|W_\Lambda^*\| \leq W_{\Lambda M} \cdot \varepsilon_\Lambda$ is the approximation error, which is upper bounded by $\|\varepsilon_\Lambda\| \leq \varepsilon_{\Lambda M}$. For simplicity, $\sigma(Y_\Lambda^T(x_k, a))$ is denoted by $\sigma_\Lambda(k)$ in the rest of the proof.

The first term $\Delta L_{\tilde{W}_{\Lambda,k}}$ can be written as

$$\begin{aligned} \Delta L_{\tilde{W}_{\Lambda,k}} &= \frac{1}{\alpha_\Lambda} \text{tr}(W_\Lambda^T(k+1) W_\Lambda(k+1)) \\ &\quad - \frac{1}{\alpha_\Lambda} \text{tr}(W_\Lambda^T(k) W_\Lambda(k)) \end{aligned} \quad (57)$$

where $\tilde{W}_\Lambda = \hat{W}_\Lambda - W_\Lambda^*$.

According to Lemma 2, the optimal cost function can be represented as

$$J^*(u, \lambda) = \sum_{j=1}^s \sum_{i=1}^L W_j^* \cdot \langle y_i, y_i' \rangle \sigma(u) + \varepsilon_c(k) \quad (58)$$

where W_c^* denotes the optimal weights of the critic network and its upper bound satisfies $\|W_c^*\| \leq W_{cM} \cdot \varepsilon_a$ is the approximation error of the costate network, which is upper bounded by $\|\varepsilon_c\| \leq \varepsilon_{cM}$. For simplicity, $\sigma(Y_c^T(x_k, a, \lambda))$ is denoted by $\sigma_c(k)$ in the rest of the proof.

The second term $\Delta L_{\tilde{W}_{c,k}}$ can be written as

$$\begin{aligned} \Delta L_{\tilde{W}_{c,k}} &= \frac{1}{\alpha_c} \text{tr}(\tilde{W}_c^T(k+1) \tilde{W}_c(k+1)) \\ &\quad - \frac{1}{\alpha_c} \text{tr}(\tilde{W}_c^T(k) \tilde{W}_c(k)) \end{aligned} \quad (59)$$

where $\tilde{W}_c = \hat{W}_c - W_c^*$.

With the weight update rule of $\hat{W}_k^c$ in (53), we can obtain

$$\begin{aligned} &W_c(k+1) \\ &= W_c(k) + \Delta W_c(k) \\ &= W_c(k) - \alpha_\Lambda \frac{\partial L_k^c}{\partial e_k^c} \frac{\partial e_k^c}{\partial \hat{J}_k} \frac{\partial \hat{J}_k}{\partial \hat{W}_k^c} \\ &= W_c(k) - \alpha_\Lambda \sigma_c(k) [\hat{J}(u, \lambda) - (U(u) + \gamma \hat{J}(\hat{u}_{k+1}, \hat{\phi}_{k+1}))]. \end{aligned} \quad (60)$$

According to the property of the Bellman equation, the cost function estimated by the BiLSTM network can be written as

$$\partial J^*(u_t, \phi_t) - U(u_t) - \gamma J^*(u_{t+1}, \phi_{t+1}) = 0. \quad (61)$$

According to Theorem 2, the control law remains admissible with the iteration process, and therefore, it is given by

$$J^*(x_{k+1}, u_{k+1}, \phi_{k+1}) < J^*(x_{k+1}, u'_{k+1}, \phi'_{k+1}) \quad (62)$$

where u'_{k+1} represents the improved policy and $\Lambda'_{k+1} = \Lambda(x_{k+1}, u'_{k+1})$. Substituting (62) into (61), the utility function $U(x_k, u_k)$ can be expressed as

$$J^*(x_k, u_k, \phi_k) - \gamma J^*(x_{k+1}, u'_{k+1}, \phi'_{k+1}) - \varepsilon_{\Delta u}^J = U(x_k, u_k) \quad (63)$$

where $\varepsilon_{\Delta u}^J$ is upper bounded by $\varepsilon_{\Delta u}^J \leq \|\varepsilon_{\Delta u, M}^J\|$.

Then, from (60) and (63), then

$$\begin{aligned} U(x_k) &= J^*(x_k, \lambda) - \gamma J^*(x_{k+1}, u_{k+1}, \phi_{k+1}) \\ &= \sum_{j=1}^s \sum_{i=1}^L W_j^* W_y^* \cdot \langle y_i, y_i' \rangle \sigma(u) + \varepsilon_c(k) \\ &\quad - \gamma \sum_{j=1}^s \sum_{i=1}^L W_j^* W_y^* \cdot \langle y_i, y_i' \rangle \sigma_{c,k+1} - \gamma \varepsilon_{c,k+1} - \varepsilon_{\Delta u}^J \\ &= W_s^{*T} \Delta \sigma_{t,t+1}^s + \varepsilon_{t,t+1}^s - \varepsilon_{\Delta u}^J \end{aligned} \quad (64)$$

where $\Delta \sigma_{k,k+1}^c = \sigma_c(k) - \gamma \sigma_{c,k+1}$ and $\varepsilon_{k,k+1}^c = \varepsilon_c(k) - \gamma \varepsilon_{c,k+1}$ is assumed to be upper bounded by $\|\varepsilon_{k,k+1}^c\| < 2\varepsilon_{cM}$.

Substituting (64) into (60), an equation is obtained

$$\begin{aligned} W_s(k+1) &= W_s(k) - \alpha_s e_k^s \sigma_s(k) = W_s(k) \\ &\quad - \alpha_s \sigma_s(k) \times [\hat{W}_s^T(k) \sigma_s(k) - (U(x_k) + \gamma \hat{W}_s^T(k) \sigma_{s,k+1})] \\ &= -\alpha_s [\sigma_s(k) W_s^T(k) \Delta \sigma_{k,k+1} - \sigma_s(k) \varepsilon_{k,k+1}^s + \sigma_s(k) \varepsilon_{\Delta u}^J] \\ &\quad + W_s(k). \end{aligned} \quad (65)$$

Assume that $\|\sigma_c(k)\| \leq \sigma_{cM}$, and then, $\|\Delta \sigma_{k,k+1}^c\| \leq 2\sigma_{cM}$ is achieved. For simplify, $\varepsilon_{k,k+1}^c - \varepsilon_{\Delta u}^J = \varepsilon_J$.

Substituting (65) into $\Delta L_{\tilde{W}_s, k}$, the following expression achieves:

$$\begin{aligned} \Delta L_{\tilde{W}_s, k} &= \text{tr}[-2\xi_{s,k} W_s^T(k) \Delta \sigma_{k,k+1}^s + 2\xi_{s,k} \varepsilon_J] \\ &\quad + \alpha_s \xi_{s,k} \xi_{s,k}^T (\Delta \sigma_{k,k+1}^s)^2 + \alpha_s \sigma_s^2(k) \varepsilon_J^2 \end{aligned} \quad (66)$$

where $\xi_{s,k} = \tilde{W}_s^T(k) \sigma_s(k)$. Since $\|\varepsilon_{k,k+1}^c\| \leq 2\varepsilon_{cM}$ and $\varepsilon_{\Delta u}^J \leq \|\varepsilon_{\Delta u, M}^J\|$, we have $\varepsilon_J \leq 2\varepsilon_{cM} + \varepsilon_{\Delta u, M}^J = \|\varepsilon_{JM}\|$.

Then, applying the Cauchy-Schwarz inequality and Young's inequality $2a^T b \leq a^T a + b^T b$, it can be represented as follows:

$$\begin{aligned} \Delta L_{W_s, k} &\leq \|\xi_{s,k}\|^2 (\|\varepsilon_{JM}\|^2 + (4\alpha_s \|\varepsilon\|^2 - \alpha_s \|\varepsilon_{JM}\|^2 - 1)) \\ &\quad + \|\varepsilon\|^2 (\alpha_s \|\varepsilon_{JM}\|^2 - 4\alpha_s \|\varepsilon\|^2 - 4\|W_s^T(k)\|^2). \end{aligned} \quad (67)$$

The above equation is negative, which means that all the output control variables will be converted to a small error boundary and the stability of the ConFrame proof completed.

IV. EXPERIMENTAL IMPLEMENTATION AND VERIFICATION

In this part, the proposed ConFrame is used for the deployment of the intelligent mining cone platform. The experiment is built with Pytorch and Linux operation system platform, MATLAB 2022a for the optimization control tasks, a core i9

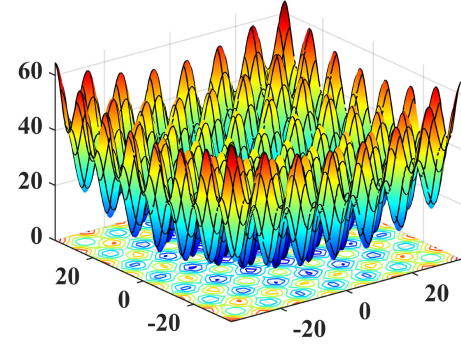


Fig. 4. Optimization global point by the proposed ConFrame framework and algorithms.

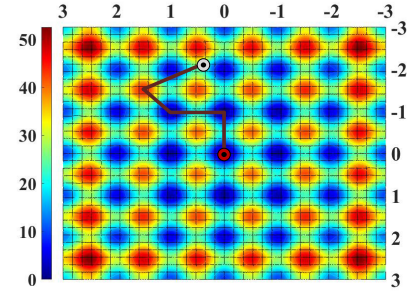


Fig. 5. Illustration of the proposed ConFrame framework for the Rastrigin optimization.

TABLE II
EVALUATION INDEXES FOR THE DIFFERENT COMPETITIVE PREDICTIONS

Methods	RMSE	MAE
RNN	0.6712 ± 0.0124	0.3488 ± 0.0082
GRU	0.4634 ± 0.0078	0.2499 ± 0.0345
LSTM	0.7533 ± 0.0030	0.5119 ± 0.0079
SVR	0.8133 ± /	0.6003 ± /
Ours	0.2234 ± 0.0021	0.1731 ± 0.0013

with RTX5000 with 128 GB memory. To land such a platform and intelligent architecture, some necessary preparation is given as follows.

A. Case Studies

1) *Numerical Case:* The first illustration case is from the numerical case. The Rastrigin function is used as the bench case to verify our proposed framework (see Fig. 4). The Rastrigin function is defined by

$$f(\mathbf{x}) = A + \sum_{i=1}^n [x_i^2 - A \cos(2\pi x_i)] \quad (68)$$

where $A = 10$ is the constant and $\mathbf{x} \in [-5.12, 5.12]^n$ is a vector with n dimension. The Rastrigin function has many local minima, so the optimization algorithms must be tested from multiple starting points. The 2-D plot of the Rastrigin function reveals the global minimum in a low valley around (0, 0) with many local minima evenly distributed through the search space. Benchmarks often test a high number of dimensions to increase the computation difficulty. As shown in many normal examples, we consider the bi-dimensional case where $n = 2$.

As shown in Figs. 5 and 6, compared to the genetic algorithm (GA)-based method, the comparative experiment

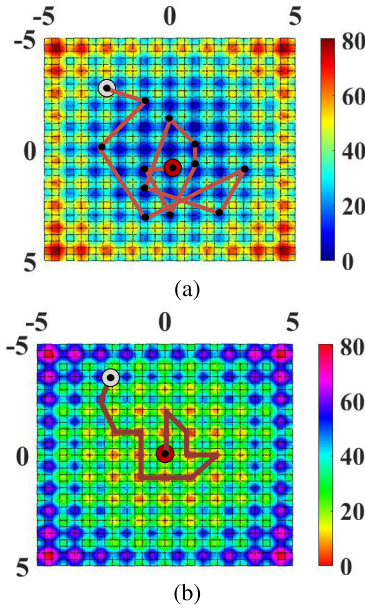


Fig. 6. Two comparative cases for the Rastrigin bench optimization. (a) PSO. (b) GA.

TABLE III

EVALUATION INDEXES FOR THE DIFFERENT COMPETITIVE CONTROL

Methods	Convergence epochs	Training time (s)
GA	12	359.45
PSO	33	428.91
Ours	5	445.71

shows that the optimization of our ConFrame finds the global optimization faster under the constraint of ranges. Fig. 6(a) describes the PSO algorithm for finding the global minimum for the Rastrigin bench function, and Fig. 6(b) gives the GA results, respectively. Compared two figures, the GA converges and finds the global minima at the iterative of 11 iterative, whilst the PSO finds it at least 13 epochs, the convergence efficiency is slower than our proposed algorithm, five epochs shown in Fig. 5.

2) *Practical Applicable Case Study*: In this part, the practical deep cone thicker system is used as the evaluation for the proposed framework. To make a comparative comparison, all the prediction model parameter initialization is set up the same with [13], and our iterative is set as 50. The number of particles of PSO is set as 20. For the PSO parameters setting, acceleration constant c_1 and c_2 are both equal to 2, and the maximum population is set as 100, the max iteration is 50 [20], [21], [22].

B. Result Analysis

Fig. 7 presents the illustrative results of the different prediction models for the underflow concentration. The testing process is operated with 300 samples. We can see that superior performance is achieved by our bidirectional LSTM algorithm. While other comparative three algorithms such as LSTM, GRU, and RNN has apparently higher variance (Table II), which means that the stability of the algorithm is worse than the proposed.

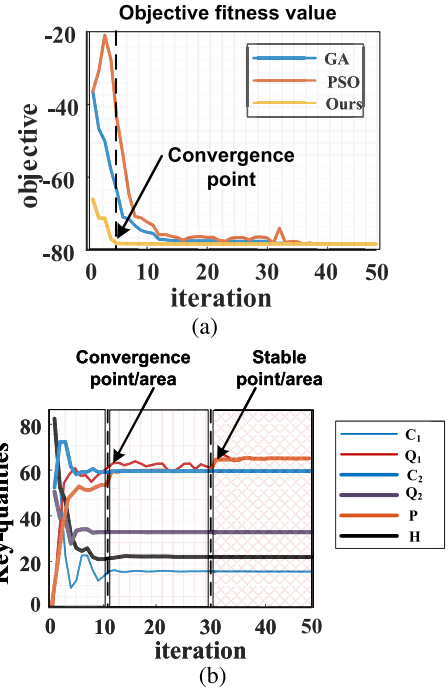


Fig. 7. Experimental evaluation. (a) Fitness values with comparative optimization. (b) Key-qualities convergence curve with increasing epochs.

Fig. 7 lists the fitness value curve with increasing epochs for the different algorithms. The PSO-based optimization and other state-of-the-art method are compared to the environments. Our presented algorithms at the earliest convergence at almost six iterative steps, and the PSO still has a higher oscillation at the stage of 33 iterations [see Fig. 6(a)]. Fig. 6(b) demonstrates the different key qualities during the production of the paste-filling industrial process. The control objective variables and the output underflow concentration converged at the point of 11, which means the system is stable and work in a balanced way (Table III). After 30 epochs of iteration, all the variables are tended to stable and the whole system operates stable and of high performance.

V. CONCLUSION

This article proposes a unified platform ConFrame for the intelligent underground paste-filling process. The presented framework implements the advanced prediction and control of the deep cone thickener. To the best of our knowledge, we first propose such a unified prediction and control platform in the underground paste-filling industry. The whole deployment promotes the development of the underground mining industry. In addition, the proposed framework addresses the challenge of high safety hazards, low mining efficiency, and high cost. The evaluation index and numeric verification of the presented work have been set as the benchmark and outperformed other state-of-the-art comparisons. The control strategy improves the interpolation ability and the robustness of the whole underground mining system and platform. It also revolts against the fast pace of the smart manufacturing industry. It promotes cutting-edge work for the development of future non-manual mining. For further consideration, it can be improved from the following aspects.

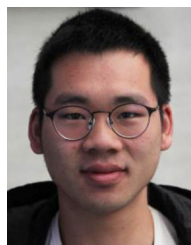
1) The abundant other algorithms can be integrated into the existing system and platform. The existing prediction

algorithm addresses the underflow concentration in DCS with high prediction accuracy, however, the whole work can be further explored with a closed-circle feedback network.

2) There is more space with the control module implementation since it can employ additional advanced control approaches. As an example, consider reinforcement learning. This related work can be reported in our future works.

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